

The Moduli of the Canonical Basis in $\text{FBL}^{(p)}[\ell_r]$: Verification and a Complete Answer to Remark 9.48

Research verification note

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Abstract

The automatic-solver packet makes a correct reduction to an absolutely p -summing norm, but then substitutes the wrong parameter into Garling's diagonal theorem. For $2 < r, p < \infty$, the correct exponent is independent of p : if

$$\frac{1}{t} = \frac{1}{r} + \frac{1}{2},$$

then the sequence $(|\delta_{e_k}|)$ in $\text{FBL}^{(p)}[\ell_r]$ is equivalent to the unit vector basis of ℓ_t . This follows directly from Garling's Theorem 9(v). A second proof is included: its lower estimate uses all Rademacher sign vectors and Khintchine's inequality, while its upper estimate follows from the already known $p = 2$ case. We also record the resulting classification for every finite free-convexity parameter.

1 Source problem and status

The source paper is T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, arXiv:2210.00614. Remark 9.48 states that for $r \geq 2$ and $1 \leq p \leq 2$, the moduli of the ℓ_r basis in $\text{FBL}^{(p)}[\ell_r]$ are equivalent to the ℓ_s basis with $1/s = 1/r + 1/2$, and asks what happens for $r, p \in (2, \infty)$.

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Remark 9.48. Suppose $r \geq 2$ and $1 \leq p \leq 2$. Corollary 9.47 implies that the moduli of the ℓ_r basis in $\text{FBL}^{(p)}[\ell_r]$ and in $\text{FBL}[\ell_r]$ are equivalent; by [20], both are equivalent to the ℓ_s basis, with $1/s = 1/r + 1/2$. We do not know what the span of these moduli is for $r, p \in (2, \infty)$.

On the other hand, it follows immediately from Proposition 9.30 that:

Corollary 9.49. *Suppose E, F are infinite dimensional Banach spaces, $p \in [1, \infty]$, $q \in (2, \infty]$, and $p \neq q$. Then $\text{FBL}^{(p)}[E]$ is not lattice isomorphic to $\text{FBL}^{(q)}[F]$.*

Remark 9.50. By Corollary 9.25, if E^* has finite cotype, then $\text{FBL}[E]$

Figure 1: Remark 9.48 of the source paper, giving the known $p \leq 2$ case and asking for $r, p > 2$.

This packet replaces the previous automatic packet of 14 June 2026. The old packet correctly reduced the question to diagonal absolutely summing operators, but used the wrong exponent from Garling's theorem. The corrected result is a full solution of the specific question in Remark 9.48. It uses Garling's 1974 diagonal theorem as an external ingredient, but the free-lattice norm reduction and the parameter identification are part of the solution.

2 Verdict on the proposed result

Let (e_k) be the canonical basis of ℓ_r , and write

$$u_k = |\delta_{e_k}| \in \text{FBL}^{(p)}[\ell_r].$$

The packet claims, for $2 < r, p < \infty$, that (u_k) is equivalent to the basis of ℓ_s with

$$\frac{1}{s} = \frac{1}{r} + \frac{1}{p}.$$

This is false. The reduction in the packet to a diagonal p -summing norm is correct; the subsequent reading of Garling's theorem is not.

Theorem 1 (Correct answer). *Let $2 < r, p < \infty$, and let*

$$\frac{1}{t} = \frac{1}{r} + \frac{1}{2}.$$

There are constants $c_{r,p}, C_{r,p} > 0$ such that every finitely supported scalar sequence (a_k) satisfies

$$c_{r,p} \left(\sum_k |a_k|^t \right)^{1/t} \leq \left\| \sum_k a_k u_k \right\|_{\text{FBL}^{(p)}[\ell_r]} \leq C_{r,p} \left(\sum_k |a_k|^t \right)^{1/t}.$$

Consequently, the closed linear span of (u_k) is isomorphic to ℓ_t . The exponent does not depend on the finite parameter p .

A quick way to see that the packet's exponent cannot be right is to take $a_1 = \dots = a_n = 1$. Theorem 1 gives

$$\left\| \sum_{k=1}^n u_k \right\|_{\text{FBL}^{(p)}[\ell_r]} \underset{r,p}{\asymp} n^{1/r+1/2},$$

whereas the proposed result predicts growth $n^{1/r+1/p}$. Since $p > 2$, the ratio of these two quantities grows like $n^{1/2-1/p}$.

3 The norm reduction is correct

Let r' be conjugate to r . For a finitely supported nonnegative sequence $a = (a_1, \dots, a_n)$, define

$$D_a : \ell_{r'} \longrightarrow \ell_1^n, \quad D_a(b) = (a_1 b_1, \dots, a_n b_n).$$

Lemma 2 (Exact diagonal reduction). *For $1 \leq p < \infty$ and $a_k \geq 0$,*

$$\left\| \sum_{k=1}^n a_k u_k \right\|_{\text{FBL}^{(p)}[\ell_r]} = \pi_p(D_a : \ell_{r'} \rightarrow \ell_1^n),$$

where π_p denotes the absolutely p -summing norm.

Proof. For $b = (b_k) \in \ell_{r'}$,

$$\left(\sum_{k=1}^n a_k u_k \right) (b) = \sum_{k=1}^n a_k |b_k| = \|D_a b\|_{\ell_1^n}.$$

Therefore the functional representation of the free p -convex lattice gives

$$\left\| \sum_{k=1}^n a_k u_k \right\|_{\text{FBL}^{(p)}[\ell_r]} = \sup \left\{ \left(\sum_{i=1}^m \|D_a b^{(i)}\|_1^p \right)^{1/p} : \sup_{x \in B_{\ell_r}} \left(\sum_{i=1}^m |\langle b^{(i)}, x \rangle|^p \right)^{1/p} \leq 1 \right\}.$$

Because $(\ell_{r'})^* = \ell_r$, this is exactly the definition of $\pi_p(D_a)$. Replacing the domain by $\ell_{r'}^n$ does not change the norm, by composing with the canonical coordinate projection and inclusion. $\square$

Thus the central question is precisely the following operator-ideal problem:

$$\pi_p(D_a : \ell_{r'} \rightarrow \ell_1) \quad \text{as a function of } a.$$

4 The external diagonal theorem

Garling uses three different parameters: the domain is $\ell_{P'}$, the range is ℓ_Q , and the summing order is a third parameter, say σ . For $Q < 2 < P$, he writes

$$\frac{1}{\phi(P, Q)} = \frac{1}{P} + \frac{1}{Q} - \frac{1}{2}.$$

His Theorem 9(v) states that, whenever

$$1 \leq Q \leq 2 < P \leq \infty,$$

the diagonal map

$$d_a : \ell_{P'} \longrightarrow \ell_Q$$

is σ -summing, for any finite summing order σ , if and only if $a \in \ell_{\phi(P, Q)}$.

Apply this with

$$P = r, \quad Q = 1, \quad \sigma = p.$$

Then

$$\frac{1}{\phi(r, 1)} = \frac{1}{r} + 1 - \frac{1}{2} = \frac{1}{r} + \frac{1}{2},$$

so $\phi(r, 1) = t$ from Theorem 1. In particular,

$$D_a \in \Pi_p(\ell_{r'}, \ell_1) \iff a \in \ell_t.$$

Garling's printed statement is a membership characterization. The uniform norm equivalence

$$\pi_p(D_a) \underset{r, p}{\asymp} \|a\|_{\ell_t} \tag{3.1}$$

follows by the closed graph theorem. Indeed, the diagonal operators form a closed subspace of the Banach operator ideal $\Pi_p(\ell_{r'}, \ell_1)$, because $\|T\| \leq \pi_p(T)$. Hence

$$X_{r, p} = \{a : D_a \in \Pi_p(\ell_{r'}, \ell_1)\}, \quad \|a\|_{X_{r, p}} = \pi_p(D_a),$$

is a Banach sequence space. Garling identifies it algebraically with ℓ_t , and coordinatewise convergence shows that the identity map has a closed graph in both directions. Formula (3.1) follows. Combining it with Lemma 2 proves the theorem for nonnegative coefficients.

Finally, the canonical basis of ℓ_r is 1-suppression unconditional. Proposition 5.3 of the source paper implies that (u_k) is 1-suppression unconditional, hence 2-unconditional, in every $\text{FBL}^{(p)}[\ell_r]$. Thus

$$\frac{1}{2} \left\| \sum_k |a_k| u_k \right\| \leq \left\| \sum_k a_k u_k \right\| \leq 2 \left\| \sum_k |a_k| u_k \right\|,$$

and Theorem 1 follows for arbitrary scalars.

5 A direct verification in the open range

Because the source paper warns that Garling's Theorem 9 contains some typographical errors, it is useful to verify the relevant conclusion without relying on the detailed case list. This also pinpoints why the exponent is $1/r + 1/2$.

Let

$$\frac{1}{v} = \frac{1}{2} - \frac{1}{r}.$$

Then $v = 2r/(r - 2)$ and $v' = t$.

Proposition 3. *Let $2 < r < \infty$, $2 \leq p < \infty$, and let $a = (a_k)_{k=1}^n$ be nonnegative. If B_p is an upper Khintchine constant, then*

$$B_p^{-1} \|a\|_{\ell_t} \leq \pi_p(D_a) \leq C_r \|a\|_{\ell_t},$$

where C_r is a constant from the already known $p = 2$ case.

Proof. For the upper estimate, the inclusion theorem for absolutely summing operators gives $\pi_p(T) \leq \pi_2(T)$ when $p \geq 2$. The known $p = 2$ case of Remark 9.48, together with Lemma 2, gives

$$\pi_2(D_a) \leq C_r \|a\|_{\ell_t}.$$

For the lower estimate, fix a nonnegative sequence $c = (c_k)_{k=1}^n$. For every sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, 1\}^n$, put

$$x_\varepsilon = (\varepsilon_k c_k)_{k=1}^n \in \ell_{r'}^n.$$

The numerator in the definition of $\pi_p(D_a)$ is

$$\left(\sum_{\varepsilon} \|D_a x_\varepsilon\|_1^p \right)^{1/p} = 2^{n/p} \sum_{k=1}^n a_k c_k.$$

The weak p -norm of the family (x_ε) is bounded, by Khintchine's inequality, as follows:

$$\begin{aligned} \sup_{y \in B_{\ell_r^n}} \left(\sum_{\varepsilon} \left| \sum_{k=1}^n \varepsilon_k c_k y_k \right|^p \right)^{1/p} &\leq 2^{n/p} B_p \sup_{y \in B_{\ell_r^n}} \left(\sum_{k=1}^n c_k^2 |y_k|^2 \right)^{1/2} \\ &= 2^{n/p} B_p \|c\|_{\ell_v}. \end{aligned}$$

The last equality is the norm formula for the multiplication operator $\ell_r^n \rightarrow \ell_2^n$ with multiplier c . Consequently,

$$\pi_p(D_a) \geq \frac{\sum_k a_k c_k}{B_p \|c\|_{\ell_v}}.$$

Taking the supremum over $c \geq 0$ and using $(\ell_v)^* = \ell_{v'}$ gives

$$\pi_p(D_a) \geq B_p^{-1} \|a\|_{\ell_{v'}} = B_p^{-1} \|a\|_{\ell_t}.$$

□

This proof shows directly that increasing the free-convexity parameter past 2 does not change the exponent. The upper estimate comes from the monotonicity $\pi_p \leq \pi_2$, while the Rademacher test forces the same ℓ_t lower estimate for every finite p .

6 Complete classification for the canonical sequence spaces

The result above combines with the known cases in the source paper to give a compact classification.

Corollary 4. *For $1 \leq r \leq \infty$, set $E_r = \ell_r$ when $r < \infty$ and $E_\infty = c_0$. Let (e_k) be the canonical basis of E_r . For every $1 \leq p < \infty$, the sequence $(|\delta_{e_k}|)$ in $\text{FBL}^{(p)}[E_r]$ is equivalent to the unit vector basis of ℓ_τ , where*

$$\frac{1}{\tau} = \min \left\{ 1, \frac{1}{r} + \frac{1}{2} \right\}, \quad \frac{1}{\infty} := 0.$$

Thus the answer is independent of every finite p .

For $p = \infty$, by contrast, $(|\delta_{e_k}|)$ is equivalent to the original canonical basis of E_r .

Proof. For $1 \leq r \leq 2$ and finite p , the moduli are equivalent to the ℓ_1 basis by Proposition 5.14 and Example 5.15 of the source paper. For $2 < r < \infty$, Theorem 1 applies; for $r = \infty$, the c_0 case is Proposition 5.10 there. The $p = \infty$ assertion follows from Proposition 6.4 of the source paper, since the canonical bases are unconditional. $\square$

7 Precisely what went wrong in the packet

1. The reduction to $\pi_p(D_a : \ell_{r'} \rightarrow \ell_1)$ is correct.
2. In Garling's notation, the letter controlling the domain $\ell_{P'}$ is P , while the summing order is a separate variable. One must substitute $P = r$, $Q = 1$, and summing order p . The packet instead inserted the summing order into the exponent $\phi(P, Q)$.
3. The unconditionality citation should be Proposition 5.3, not Proposition 5.2.
4. The heuristic $p \rightarrow \infty$ is misleading. For every fixed finite p the exponent is $1/r + 1/2$, whereas at $p = \infty$ the sequence is equivalent to the ℓ_r basis. The equivalence constants are not uniform as $p \rightarrow \infty$, and the order of the limits in p and in the support size matters.

8 Literature conclusion

No later paper explicitly advertised as an answer to Remark 9.48 was found in the searches carried out for this note. Nevertheless, the full answer uses a classical ingredient: Lemma 2 turns the free-lattice question into exactly case (v) of Garling's 1974 diagonal theorem. Accordingly, the corrected identification should not be presented as a new operator-ideal theorem, but it should be recorded as a full answer to the source remark.

References

- [1] D. J. H. Garling, *Diagonal mappings between sequence spaces*, *Studia Math.* **51** (1974), 129–138. <https://eudml.org/doc/217909>
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